Odd kernels and cyclotomic polyhedra: complete manuscript costs

Human Compression project

The fixed source is Hou–Feng–Leemans–Qu, *String C-groups of order* $4p^m$, arXiv:2407.10388v1, 15 July 2024. Its twelve pages are compared with our complete four-page common theory and five-page application. The declared corrected principal contract has 25 source records; it excludes the introductory numerical census and general Hughes classification. Full source length does not mean that every printed auxiliary assertion was recovered.

The whole topic and the conditional application. The primary coordinate counts NFC characters without whitespace after the accepted visible-glyph calibration and removal of exact running furniture. Titles, abstracts, proofs, tables' extractable text, references and contacts remain counted. The entire common paper is paid once within this topic.

$$R_{\text{written}} = \frac{25115}{7118 + 12744} = \frac{25115}{19862} = 1.264, \qquad r \mid (U_F, V_F) = \frac{25115}{12744} = 1.971.$$

The conditional figure assumes acquisition of the whole common paper and the declared complete validation package V_F. Its programs check the uniform written proofs; none supplies a finite premise of those proofs. Overlap with other topics gives no discount on the common paper.

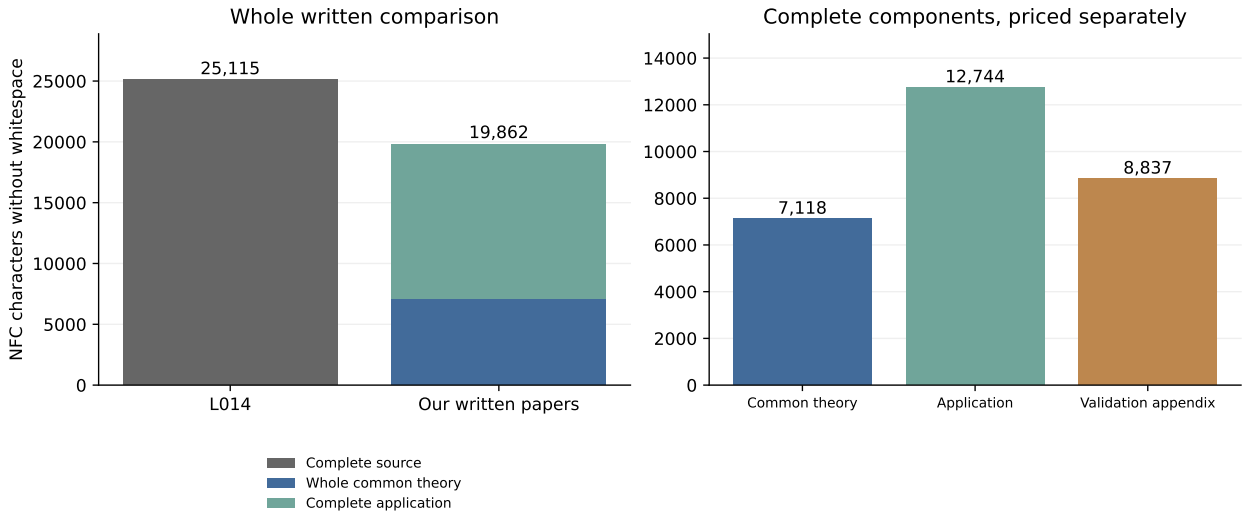
Complete document	Pages	Characters
Source L014	12	25115
Common theory U_F	4	7118
Application F1	5	12744
Validation appendix	4	8837

The complete validation cost. The four-page appendix prints all three programs, including comments and printed continuation marks. It gives the separate comparison

$$R_{\text{validation}} = \frac{25115}{19862 + 8837} = \frac{25115}{28699} = 0.875.$$

Using the three complete raw files instead costs 8061 characters, giving $25115/27923 = 0.899$. The appendix and raw files are alternative representations; they are never charged together. The two operational drivers cost 10881 characters separately. The reused Lean file costs 2269 separately and proves five scoped declarations, not the entire source contract. No fresh mathematical or formal result is asserted here.

Retained inputs and unknown costs. The tightness converse retains Cunningham–Pellicer's classification, with its exact odd/even hypotheses and uniqueness conclusion. Its full proof cost, other external proof costs, historical software and source graphics remain unknown. The 43 one-pixel source images and two-dimensional formula layout are inventoried but not priced by extracted-text metrics. These are comparisons of available texts, not total research effort or minimum proof complexity.



The left panel compares the whole source with both complete written papers. The right panel exposes each complete cost, including validation. There is only one application in this topic; these bars establish no recurring sawtooth pattern. A later dependency graph may locate reuse within the proofs without changing this full-document accounting.

Sensitivity to the unit of length. For calibrated selected text with NFC normalization, the three displayed comparisons are as follows. Different units answer different questions; the primary nonspace-character convention was fixed before these factors were evaluated.

Unit	Whole written	With validation	Conditional
Whitespace words	1.725	1.479	2.753
Alphanumeric units	1.616	1.137	2.549
Nonspace characters	1.264	0.875	1.971
Characters, collapsed spaces	1.352	0.971	2.116
UTF-8 bytes, collapsed spaces	1.251	0.936	1.989

The machine-readable calculation retains all five units under NFC and NFKC, for untrimmed raw extraction, selected raw extraction and calibrated selected extraction: thirty coordinates and 120 exact fractions. Whole raw programs, drivers and formal files retain their separate input counts. Rounded figures here are readers for the exact integer fractions, not substitutes for them.

Extraction and verification. The accepted input audit independently recounted all 120 PDF and 60 raw-file metric cells. It inspected all 25 physical pages and aligned all 25,574 nonspace source characters with their original PDF text bytes. Ninety-five occurrences repair visible glyph identity: sixteen mathematical alphabet glyphs, 78 pieces of tall square brackets and one normality sign. These repairs neither fix the source's mathematical marking nor rearrange formulas. Raw extraction remains available as a sensitivity. The pinned parsers are pypdf 6.13.0 and fontTools 4.63.0; their implementation and Unicode semantics are declared computational dependencies.

The numerical builder reads the accepted input records, checks their frozen manifests and independently reviewed counts, and forms the displayed complete denominators. The original input policy and mathematical admissions remain unchanged. Independent numerical review is required before publication.